Theorem. (Erdős #741(i)) Let $A \subseteq \mathbb{N}$ be such that $A + A$ has positive upper density. Can one always decompose $A = A_1 \sqcup A_2$ such that $A_1 + A_1$ and $A_2 + A_2$ both have positive upper density? We show the answer is yes.
Proof. We show that the answer is yes. We use an alternating block partition. Given a rapidly growing sequence $M_0 < M_1 < M_2 < \cdots$, we define

$$A_1 = A \cap \bigcup_k (M_{2k}, M_{2k+1}], \qquad A_2 = A \setminus A_1.$$

In odd-indexed intervals $(M_{2k}, M_{2k+1}]$, all elements of A belong to A_1; in even-indexed intervals $(M_{2k+1}, M_{2k+2}]$, they all belong to A_2. The sequence M is chosen to grow fast enough that each block dwarfs all previous ones. We split into two cases.

Case 1: A **has positive upper density.** There exist a constant $c > 0$ and a strictly increasing sequence of scales along which $|A \cap [1, N]| \geq c \cdot N$. Using a dependent-choice argument, we extract a rapidly growing sequence M_k such that for each k:

- $|A \cap [1, M_{k+1}]| \geq c \cdot M_{k+1}$ (density is retained at the next scale),
- $|A \cap [1, M_k]| \leq \frac{c}{4} \cdot M_{k+1}$ (the "past" is negligible relative to the "future").

Because each new block contains all the "fresh" elements of A, looking at scale M_{2k+1} shows that A_1 has positive upper density, and looking at scale M_{2k+2} shows the same for A_2. A short argument then lifts this: if a set has positive upper density, so does its sumset with itself.

Case 2: A **has zero upper density but** $A+A$ **has positive upper density.** This is the harder case, since A is too sparse to guarantee positive density for the parts directly. Instead, we argue about the sumsets themselves. Since $A+A$ has positive upper density, there exist $c > 0$ and a sequence of scales along which $|(A+A) \cap [1, N]| \geq c \cdot N$. Since A has zero upper density, $|A \cap [1, N]| = o(N)$, so for any fixed K there are arbitrarily large N where $(K+1) \cdot |A \cap [1, N]| \leq \frac{c}{4} \cdot N$. By dependent choice, we extract a rapidly growing M_k such that for each k:

- $|(A + A) \cap [1, M_{k+1}]| \geq c \cdot M_{k+1}$,
- $(M_k + 1) \cdot |A \cap [1, M_{k+1}]| \leq \frac{c}{4} \cdot M_{k+1}$.

The key ingredient is a combinatorial sumset bound: if $A = A_1 \cup A_2$ and every element of A_2 in $[1, N]$ is at most K, then

$$|(A + A) \cap [1, N]| \leq |(A_1 + A_1) \cap [1, N]| + (K + 1) \cdot |A \cap [1, N]|.$$

The idea is that any sum involving an element of A_2 has one summand bounded by K, giving at most $(K + 1) \cdot |A \cap [1, N]|$ such sums. Applying this bound at alternating scales:

- At scale $N = M_{2k+1}$: all elements of A_2 in $[1, N]$ lie below M_{2k}, so the bound with $K = M_{2k}$ gives $|(A_1 + A_1) \cap [1, N]| \geq \frac{3c}{4} \cdot N$.
- At scale $N = M_{2k+2}$: symmetrically, all elements of A_1 in $[1, N]$ lie below M_{2k+1}, giving $|(A_2 + A_2) \cap [1, N]| \geq \frac{3c}{4} \cdot N$.

Since these bounds hold for infinitely many N, both sumsets have positive upper density. $\square$